

Warm up: Distance, velocity and acceleration

Poll Question

Suppose $f(t)$ is the height of a coconut falling from a Palm tree, $0 \leq t \leq 1.8$.
Then

(a) $f'(t)$ is...

A

positive

B

negative

C

zero

(b) $f''(t)$ is...

A

positive

B

negative

C

zero

Lecture 10

The Second Derivative Marginal Cost and Revenue

Big Questions

- How can we understand the second derivative through each representation?
- Can we relate the derivative to the concepts of marginal cost and revenue from economics?

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Recall: What can a function's 1st derivative tell you about its graph?

Recall: In Lecture 7, we said that:

- if $f'(x) > 0$ on an interval, then, on that interval:

- If $f'(x) < 0$ on an interval, then, on that interval:

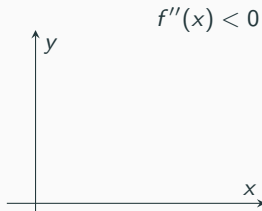
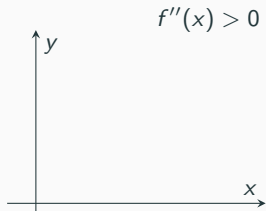
What can a function's second derivative tell you about its graph?

Poll Question

(a) Suppose that $f''(x) > 0$ on an interval. This means that, on that interval ☐ A f ☐ B f' ☐ C f'' is increasing. Hence, as we move to the right along the graph of $y = f(x)$, the ☐ A graph ☐ B slope is increasing. Therefore, the graph of $y = f(x)$ is ☐ A concave up ☐ B concave down.

(b) Suppose that $f''(x) < 0$ on an interval. By similar reasoning, this means that the graph of $y = f(x)$ is on that interval.

Sketch two pictures below to illustrate your answers.



Second derivative, according to a table

The vertical height h (in m) of a paper airplane t seconds after it is thrown into the air is given by the function $h = w(t)$ (defined by the table below).

t (in s)	100	110	120	130	140
$w(t)$ (in m)	10.7	6.3	4.2	3.5	3.3

Poll Question

(a) Does w' appear to be positive or negative? Does w'' appear to be positive or negative?

A $w' > 0$ and $w'' > 0$

B $w' < 0$ and $w'' > 0$

C $w' > 0$ and $w'' < 0$

D $w' < 0$ and $w'' < 0$

(b) What does your answer to (a) tell you about the airplane's motion?

Wrap-up on Second Derivatives: Interpretations in Four Representations

Wrap-up on Second Derivatives: Interpretations in Four Representations

Tables

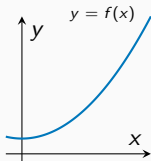
t	10	20	30	40	50
$w(t)$	3	13	28	48	78
$w'(t) \approx$	1	1.5	2	3	

Wrap-up on Second Derivatives: Interpretations in Four Representations

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Graphs

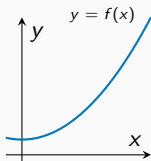


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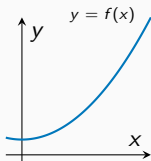
A car drives through an intersection after the stop light changes from red to green, and $f(t)$ is the distance travelled by the car t seconds after it takes off.

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Formulas

Exercise: Describe a second derivative in each of the 4 representations.

Motivation: Costs of Producing Guitars

Poll Question

The cost in dollars of producing q guitars is given by $C = f(q)$. How would you interpret the statement $f'(4000) = 150$? Choose the best answer.

- A** If we are currently producing 4000 guitars, the cost of producing a few more guitars is about \$150 per guitar.
- B** If we are currently producing 4000 guitars, the average cost of production is \$150 per guitar.
- C** The cost of producing a few more guitars is about \$150 per guitar.
- D** None of the above.

Marginal Cost

Definition

If the cost of producing q units of a good is $C(q)$, the **marginal cost** is defined to be

$MC(q) =$

Discussion Question

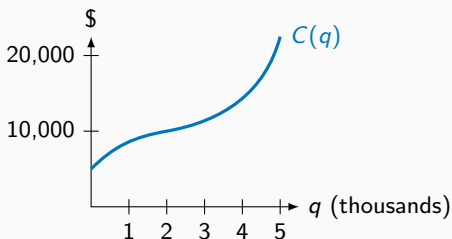
Olivia notices that her ECO textbook has the definition on the right. “It seems like ECO and MAT have completely different definitions of marginal cost, right?”

The **marginal cost** of producing a good or service is the additional cost incurred by producing one more unit of that good or service.

How might you reply to Olivia? Please discuss with your pod.

Producing Pendants

Graphed below is the cost function for producing amethyst pendants.



Poll Question

- (a) Does it cost more to produce the **A** 500th or **B** 2000th pendant?

Note. The cost of producing the 500th pendant (*i.e.* pendant #500), is not the cost of producing the first 500 pendants.

- (b) At roughly which production level q (in thousands) is the marginal cost minimized? **A** 0 **B** 1 **C** 2 **D** 3.5

Motivation for next time: Telephones

To produce q telephones, a manufacturer's cost is $C(q)$. To sell q telephones, the revenue is $R(q)$. Suppose that:

$$C(300) = 5200,$$

$$R(300) = 8400,$$

$$C'(300) = 13,$$

$$R'(300) = 8.$$

Discussion Question

How much **profit** does the company make by selling 300 telephones?

Poll Question

The company is currently producing and selling 300 telephones. In order to generate more profit, they should...



increase



decrease



not change

their sales. Assume they sell everything they produce.

To do before next class:

- Read Section 2.5.
- Complete #11-17 of Homework 3.
- Complete questions 4-6 from the Week 3 Checkpoint Workbook.
- Start working on your First Draft for Project 1.

Have a great day!

Definition

Suppose that cost, revenue and profit of producing q units of a good is

$$C(q), R(q) \text{ and } \pi(q)$$

respectively. We define the **marginal cost**, **marginal revenue**, and **marginal profit** as follows.

- The marginal cost is defined as
- The marginal revenue is defined as
- Similarly, the marginal profit is

Poll Question

How are marginal profit, marginal cost and marginal revenue related to each other?

A $MP = MR + MC$

B $MP = MR' + MC'$

C $MP = MR - MC$

D $MP = MR' - MC'$

E None of the above

Explain in non-technical terms.